Oliver Lammel[1]
e-mail: oliver.lammel@dlr.de

Harald Schütz

Guido Schmitz

Rainer Lückerath

Michael Stöhr

Berthold Noll

Manfred Aigner

German Aerospace Center (DLR),
Institute of Combustion Technology,
Pfaffenwaldring 38-40,
70569 Stuttgart, Germany

Matthias Hase

Werner Krebs

Fossil Power Generation Division,
Energy Sector,
Siemens AG,
45473 Mülheim an der Ruhr, Germany

FLOX® Combustion at High Power Density and High Flame Temperatures

In this contribution, an overview of the progress in the design of an enhanced FLOX® burner is given. A fuel flexible burner concept was developed to fulfill the requirements of modern gas turbines: high specific power density, high turbine inlet temperature, and low NO_x emissions. The basis for the research work is numerical simulation. With the focus on pollutant emissions, a detailed chemical kinetic mechanism is used in the calculations. A novel mixing control concept, called HiPerMix®, and its application in the FLOX® burner are presented. In view of the desired operational conditions in a gas turbine combustor, this enhanced FLOX® burner was manufactured and experimentally investigated at the DLR test facility. In the present work, experimental and computational results are presented for natural gas and natural gas + hydrogen combustion at gas turbine relevant conditions and high adiabatic flame temperatures (up to $T_{ad} = 2000$ K). The respective power densities are $P_A = 13.3$ MW/m^2 bar (natural gas (NG)) and $P_A = 14.8$ MW/m^2 bar (NG + H_2), satisfying the demands of a gas turbine combustor. It is demonstrated that the combustion is complete and stable and that the pollutant emissions are very low. [DOI: 10.1115/1.4001825]

Keywords: FLOX®, HiPerMix®, flameless combustion, gas turbine combustion, high temperature gas turbine, fuel flexibility, hydrogen combustion, numerical simulation, high pressure test rig, laser diagnostics

1 Introduction

The concept of lean premixed combustion in stationary gas turbines has led to a significant reduction in nitrogen oxide emissions in comparison to nonpremixed gas turbine combustion [1–3]. However, its liability to thermoacoustic instabilities, the risk of flashback, and its sensitivity with respect to changes in the fuel composition are limitations for a safe and reliable operation [4–7]. To meet future goals of power generation, which include low NO_x and CO emissions at high gas turbine efficiency and especially high fuel flexibility due to the need to burn fuels stemming from all kind of sources and CO_2 sequestration, the potential of new burner concepts is investigated and evaluated. Here, the FLOX® [8] combustion technology, also termed flameless, volume, or mild combustion [9–12], presents an attractive alternative for low-emission combustors.

The term FLOX® is an acronym for "flameless oxidation." The initial atmospheric FLOX® design was characterized by a nonvisible flame [10]. The reaction zone was widely distributed. Therefore, the intensity of the visible chemiluminescence in a furnace with hot luminous walls went below the visibility limit. Operated at high pressures, the flame in a FLOX® combustor with optical access is observable.

The presented burner using the FLOX® principle is characterized by high momentum jets of a partially premixed fuel air mixture, which are discharged through orifices arranged on a circle (see Fig. 4). The axial high momentum jet flow provides intense mixing, which reduces peak temperatures in the downstream flame relevant for high NO_x emissions (Fig. 1). High flashback resistance is obtained through the absence of low velocity zones in the applied burner concept, which favors this concept for multi-fuel application requested in next generation gas turbines.

A first result from numerical simulation is shown in Fig. 1, where a longitudinal sectional drawing through the combustor axis and two opposite jet nozzles is displayed to demonstrate the FLOX® principle. The streamlines represent the flow, and we recognize a strong inner recirculation zone. In view of the pollutant formation, the molar fraction distribution of CO and NO is shown on the upper and the lower half of Fig. 1, respectively. At first glance, we observe that the reaction zone is far detached from the nozzle exit.

The FLOX® combustor investigated at the test facilities of the DLR Institute of Combustion Technology, in conjunction with numerical simulations, is a can combustor with 12 individual jet nozzles equidistantly arranged on a concentric circle on the combustor dome. For burner nozzle diameters of 4.5 mm and 6 mm, it could be demonstrated in a previous work [13–15] that the FLOX® concept performs very well in a wide operational range for pressure (up to 30 bars) and air to fuel excess ratio (up to $\lambda = 3.0$) as well. If the gas velocity at the burner nozzle exit is limited to $v_{\text{nozzle}} < 150$ m/s by reason of the burner pressure loss, the specific thermal power density for natural gas combustion is about $P_A = 2.6$ MW/m^2 bar (related to the dome area) or $P_V = 13.2$ MW/m^3 bar (related to combustor volume), respectively. However, in a gas turbine combustor, a value of at least $P_A = 10$ MW/m^2 bar is required. Accordingly, the dump ratio (dome area related to the accumulated cross sectional area of the burner nozzles) has to be decreased to a value of 6.4 corresponding to a burner nozzle diameter of 12 mm. In this case, the original FLOX® burner does not comply with the above-mentioned requirements of a "FLOX®-type combustion" resulting in stability problems and elevated pollutant emissions.

With the aid of numerical simulations, a novel burner was de-

[1]Corresponding author.

Contributed by the International Gas Turbine Institute (IGTI) of ASME for publication in the JOURNAL OF ENGINEERING FOR GAS TURBINES AND POWER. Manuscript received April 8, 2010; final manuscript received April 13, 2010; published online August 25, 2010. Editor: Dilip R. Ballal.

Copyright © 2010 by ASME

Downloaded 15 Jan 2011 to 129.137.89.240. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

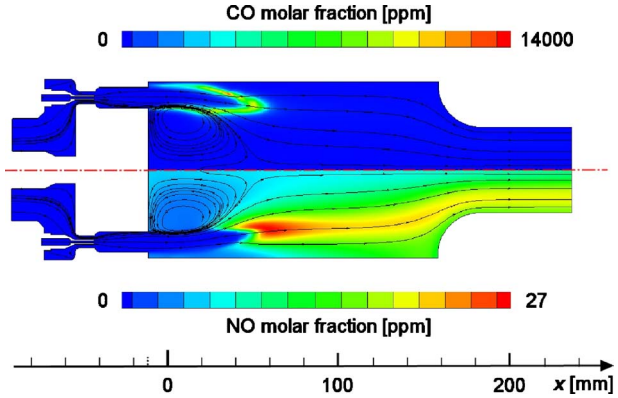

Fig. 1 Enhanced FLOX® burner with high power density operated with natural gas at 7 bars with a high adiabatic flame temperature of 1996 K. Results from numerical simulation. Flow field indicated with streamlines. Molar fraction distribution of CO at the top and of NO at the bottom.

veloped, which implements HiPerMix® [16] in the FLOX® burner concept. The characteristic feature of this HiPerMix® burner is the distinct control over the mixing performance.

2 Numerical Simulations

2.1 CFD Code and Combustion Model. The novel burner presented in this paper is completely designed on the basis of numerical simulations carried out by the DLR Institute of Combustion Technology. For this purpose, the BLUEFLAME code is applied. BLUEFLAME is a newly realized F90 finite volume computational fluid dynamics (CFD) code based upon the material volume formulation of the balance equations for compressible fluids. In its basics, the numerical solution procedure is similar to the arbitrary Lagrangian–Eulerian (ALE) method developed by Hirt et al. [17] and Pracht [18], which is also used in a series of KIVA codes [19]. The Lagrangian approach implies a truly unsteady solution procedure where the finite mesh volumes move with the fluid under the constraint that there is no convective flux between the computational cells. Inherent to this formulation is the fact that the Navier–Stokes equations are not explicitly nonlinear in the velocity and that they are not explicitly coupled with the species mass transport equations by convective terms. As a consequence, the species densities may be sequentially computed within each time step in a decoupled mode from the Navier–Stokes solution. This particular feature allows the trouble-free incorporation of complex chemistry mechanisms with any arbitrary number of species and reactions into the simulation of combustion processes. Furthermore, numerical experiments demonstrate a robust and accurate solution behavior in a wide range of Mach numbers between almost 0 and 25 of hypersonic flow. Finally, turbulence is accounted for by the "traditional" Reynolds averaging of the unsteady transport equations (URANS) supplemented by the k-ε equations. This approach is a simple representation of the turbulent length scale spectrum by an integral measure. In this regard, it seems to be physically more appropriate to directly simulate the large eddies by a solution of transport equations and to model only the scales not resolved by the computational mesh. This hybrid procedure is known as the very large eddy simulation (VLES) model [20] also available in the BLUEFLAME code.

The data structure underlying the BLUEFLAME code is on the basis of polygonal face elements with either two associated polyhedral cell elements if the face is an internal face or only one associated cell if the face is external. This makes the code extremely flexible with respect to its application to geometrically complex configurations. The finite approximation is of second order in space and time. All the physical quantities are discretized in the centers of the polyhedral cell elements that constitute the control volumes in the finite volume approach. In particular, the described data structure allows the use of computational meshes from any mesh generator and gives easy access to the dynamic local mesh adaption as well as to the dynamic load balancing in the context of parallel computing. These are very important aspects in the practical application of the code to real combustor configurations.

In the present paper, a computation of combustion is performed using the combustion code IGNIS This is a powerful chemistry simulation package, which includes different turbulent combustion models. IGNIS may be coupled with any arbitrary CFD code via user-defined functions supplying the internal energy and species densities transport equations with the appropriate source terms. If coupled with BLUEFLAME, the communication is via F90 modules. Within one computational time step, each cell is considered as a closed reactor where the combustion is independently computed likewise under isochoric or isobaric conditions. The code kernel is a highly optimized F90-routine solving complex chemical systems of elementary Arrhenius-type reactions for the production and consumption rate of the involved species. The associated matrix of the nonlinear system of equations is very "stiff," covering characteristic chemistry times of fastest and slowest reactions in a typical range of $10^{-9}-10^{6}$ s. Due to this stiffness, the fully implicit solution of the matrix equation applying a Newton–Raphson procedure is difficult and may fail for certain compositions. On the other hand, the fully explicit solution though very simple is strongly time step dependent and frequently leads to an unphysical solution (negative species densities). A feasible compromise between implicit and explicit solution procedures is a semi-implicit solution procedure according to the so-called "reference species technique" [19], which in all our simulations has shown a very robust solution behavior. Here, the basic idea is to linearize the nonlinear matrix only with respect to the species, which for each elementary reaction is in greatest danger of being driven negative. In the IGNIS code, chemistry is first solved within each computational cell using the implicit Newton–Raphson procedure. If this fails to converge, the reference species technique, which always yields a result, is applied.

From a previous work [14], we know that the FLOX® combustion is notably dominated by finite rate chemistry effects. With our focus on the pollutant emission, we decided to use the complex GRI3.0 reaction mechanism comprising 325 elementary reactions and 53 species for the natural gas combustion. In practice, the solution of a very complex chemistry system is prohibitive for the concurrent use of an elaborate turbulence/chemistry interaction model (for example, the presumed probability density function (pdf) approach) by reason of the computational cost. Therefore, the simple partially stirred reactor (PaSR) model [14,21] is applied in the present computations, which is similar to the eddy dissipation concept (EDC) model [22].

2.2 The HiPerMix® Burner. As stated in Sec. 1, it was necessary to increase the specific thermal power density of an enhanced FLOX® burner by at least a factor of 4 to fulfill the requirements of modern gas turbines. This led to a scale-up of the jet nozzles to a diameter of 12 mm, which was accompanied by stability problems and elevated pollutant emission. The reason is that the axially injected fuel jet is not sufficiently mixed with the surrounding air prior to combustion. Consequently, a new burner has to be developed with improved mixing performance.

The result of this development is the HiPerMix® (high performance mixing) burner [23] designed within an evolutionary process of numerical simulations. Basically, the burner consists of a rectangular inlet channel (annular clearance) passing into a circular mixing pipe. A schematic of the burner is displayed in Figs. 2 and 3.

Channel and pipe axes are coincident, constituting the burner axis. The fuel is axially injected in line with the burner axis at the transition between channel and mixing pipe (see Fig. 2(*c*)). It is essential that the channel height is smaller than the diameter of the

Downloaded 15 Jan 2011 to 129.137.89.240. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

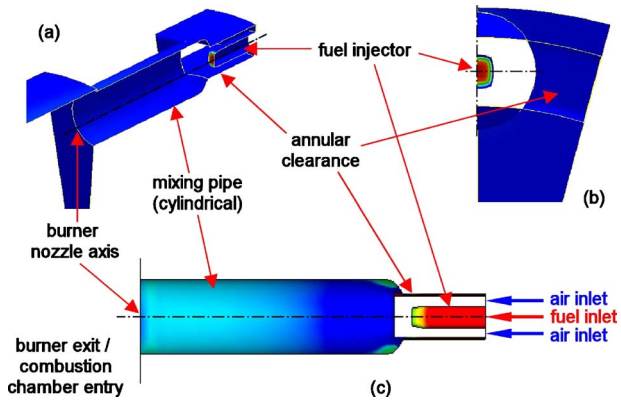

Fig. 2 Schematic of the HiPerMix® burner: (*a*) perspective view, (*b*) cross sectional slice, and (*c*) longitudinal slice

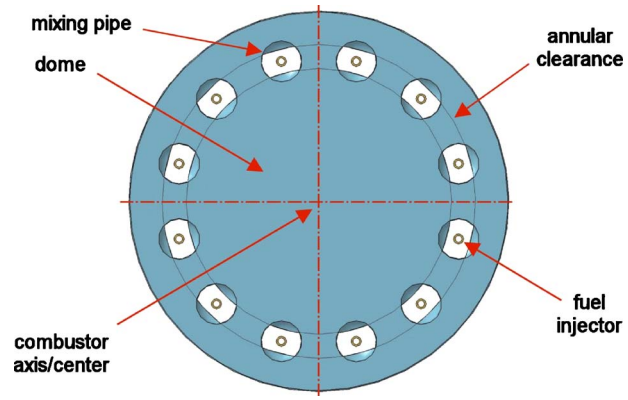

Fig. 4 Dome of a can combustor with 12 HiPerMix® burner nozzles equidistantly mounted on a concentric circle around the combustor axis

mixing pipe whereas the channel width is larger. The consequence of this is the formation of four counter-rotating axial vortices in the mixing pipe. The center lines of these vortices are parallel and symmetric to the mixing pipe axis. In this way, the four vortices dilute the fuel jet and distribute the fuel in transverse direction. The strength of the secondary vortices, which proportionally affects the mixing performance, depends on the aspect ratio of height to width of the inlet channel. For clarification, the secondary flow on a cross sectional slice in the mixing pipe is visualized by a vector plot in Fig. 3(*b*). A perspective impression of the flow is given by the streamline representation in Fig. 3(*c*). Furthermore, the color distribution on the slices is a measure for the local fuel concentration (see the color legend of Fig. 3).

We observe that in the case of a quadratic channel (Fig. 3(*a*)), the secondary flow is almost zero and the initially high fuel concentration is largely preserved downstream in the mixing pipe. In contrast to this, the fuel is essentially more distributed at the same axial position if the aspect ratio of channel height to width is, for example, chosen to be 1/2. On the one hand, this is due to the convective transport of the fuel by the secondary vortices. On the other hand, the appearance of the secondary vortices further increases the turbulent dispersion of the fuel by amplification of the turbulence production processes. However, it should be kept in mind that the improvement of the turbulent transport is always accompanied by an increase in the pressure loss due to turbulent dissipation.

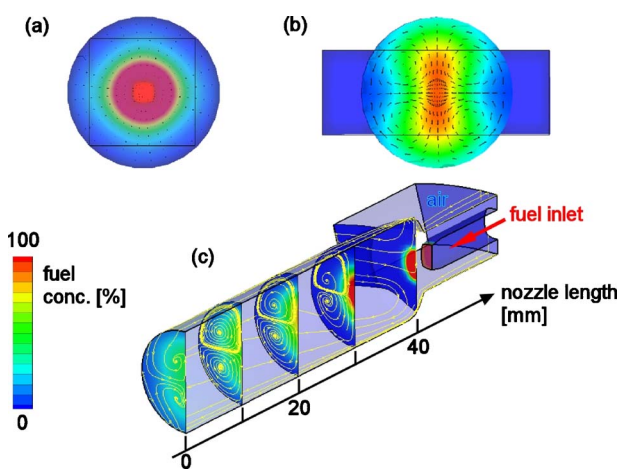

Fig. 3 Computational result of flow and fuel concentration distribution (fuel concentration (%)) in the HiPerMix® burner. (*a*) Quadratic inlet channel, (*b*) rectangular inlet channel (aspect ratio height to width=1/2), and (*c*) perspective streamline representation.

Summing up, we may state that the mixing performance of the HiPerMix® burner is reliably and steplessly variable to meet the fuel distribution requirements of a stable and low-emission combustion. So far, we have illustrated the effectiveness of the HiPerMix® concept. A schematic of its use in a can combustor investigated at our test facilities is shown in Fig. 4, where 12 burners are equidistantly mounted on a concentric circle around the combustor axis.

3 High Pressure Tests

The high pressure tests carried out during the development of an enhanced FLOX® burner were performed at the high pressure combustion test rig of the DLR Institute of Combustion Technology in Stuttgart (HBK-S). Extensive combustion diagnostic techniques were applied for the basic characterization of the combustor's operation and for the measurement of validation data sets for the numerical simulation.

3.1 High Pressure Test Rig and Combustor. The high pressure combustion test rig HBK-S is used for the investigation of scaled combustors at gas turbine relevant conditions. It is characterized by an extensive optical access for laser diagnostics (eight planar high pressure windows in four radial directions in two axial positions with a size of $160 \times 140\ \text{mm}^2$ of each window). In addition, a conventional exhaust gas analysis is available (ABB Limas 11, ABB URAS 14, and ABB Magnos 16). The test rig can be operated with pressures up to 40 bars and provides a maximum air mass flow of 1.3 kg/s, which can be electrically preheated up to 1000 K. The fuel supply allows experiments with different gaseous and liquid components (natural gas, with propane and butane admixtures, kerosene, and synthesis gases containing hydrogen, carbon monoxide, and nitrogen). Additional information can be found in Ref. [24].

Inside the high pressure housing, a test carrier is used, consisting of a supporting frame, elements for air flow guidance, an additional air preheater close to the burner, and the combustor itself. The test carrier possesses numerous temperature and pressure instrumentation locations.

The combustion chamber provides optical access through six quartz glass windows (Fig. 5). Its hexagonal shape is a compromise between the desired cylindrical form for the can liner and plane windows for the application of laser diagnostic techniques. The dimensions of the combustion chamber are an axial length of 170 mm (burner front plate to exhaust gas nozzle), a radial window distance of 100 mm, and a frame distance of 111 mm. The exhaust gas nozzle has an outlet diameter of 50 mm.

The enhanced FLOX® burner consists of 12 single nozzles in one nozzle row on a pitch circle diameter of 84.5 mm. The details

Downloaded 15 Jan 2011 to 129.137.89.240. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

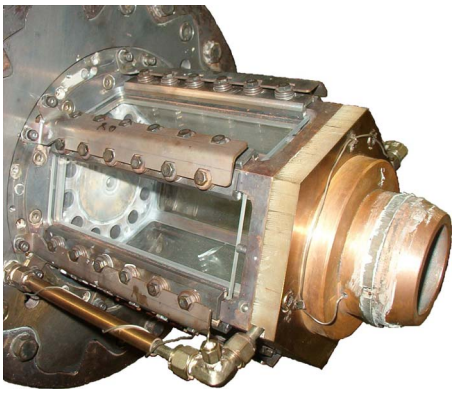

Fig. 5 Combustor for high pressure test rig HBK-S. Enhanced FLOX® burner, version 2A, and hexagonal combustion chamber with quartz glass walls for optical access.

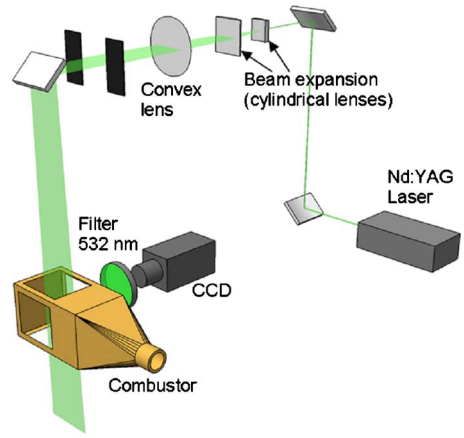

Fig. 6 Schematic of setup for PIV measurements

of the burner's fuel injection (HiPerMix® nozzle design) are described in the previous section. The flames are ignited by a small additional burner, which is located in the center of the nozzle arrangement. This ignition burner is switched off during the operation of the FLOX® burner. The burner front plate is cooled by impingement.

During the experiments, four different versions of the burner were investigated. Results for the subversions 2A and 2D are shown in this paper. The nozzle length for both combustors is 28 mm. The enhanced FLOX® burner, version 2A, has a nozzle diameter of 12 mm (dump ratio of 6.4). The nozzle diameter of version 2D is decreased to 10 mm; therefore, the dump ratio rises from 6.4 to 9.2.

3.2 Operating Conditions. As described in the previous sections, the enhanced FLOX® burner with 12 mm HiPerMix® nozzles is designed on the basis of numerical simulation for an operating point of pressure $p=7$ bars, nozzle exit velocity $v_{\text{nozzle}}=90$ m/s, air preheat temperature $T=673$ K, and equivalence ratio $\lambda=1.6$ with natural gas as its fuel. The calculated value for adiabatic flame temperature at that point is 1723°C (1996 K), and the area specific power density is $P_A=13.3$ MW/m^2 bar.

For the characterization of the burner in the high pressure test rig, different operating conditions were applied. Air to fuel excess ratio sweeps were performed at different pressure levels, nozzle exit velocities, air preheat temperatures, and different fuel compositions, respectively. An overview is given in Table 1.

Table 1 Operating condition overview. Range or maximum value for pressures p, nozzle exit velocities v_{nozzle}, air preheat temperature T, air mass flows Q_{air}, natural gas mass flows Q_{NG}, H_2 and C_3H_8 fuel content f, and thermal power (absolute value P, per area pressure P_A, and per volume pressure P_V) for enhanced FLOX® burner versions 2A and 2D

	2A	2D
$\varnothing_{\text{nozzle}}$ (mm)	12	10
p (bar)	5–7	7
v_{nozzle} (m/s)	70–150	90–133
T (K)	673–773	673
Q_{air} (g/s)	<660	<460
Q_{NG} (g/s)	<26	<19
$f(H_2)$ (vol %)	0–50	0–100
$f(C_3H_8)$ (vol %)	0	0–20
P (MW)	0.6–1.3	<1
P_A (MW/m^2 bar)	10–23	<16
P_V (MW/m^3 bar)	53–117	<81

3.3 Measurement Methods Overview. In a first step, basic combustion diagnostic techniques were applied. For each operating point, the size and position of the flame zone were observed with OH* chemiluminescence (OH*-CL). The concentrations of O_2, CO, CO_2, NO, and NO_2 in the exhaust gas were recorded continuously. Dynamic pressure in the combustion chamber was determined with piezoelectric pressure transducers and was evaluated by fast Fourier transformation (FFT) once per second.

In addition, extensive data sets were measured for selected operating conditions, which can be used for a detailed view and understanding of the combustion behavior and for the quantitative validation of numerical simulations. In this context, the composition of the natural gas was analyzed, and flow fields and temperatures were measured with laser diagnostic techniques.

3.4 OH* Chemiluminescence Imaging (OH*-CL). The signal intensity of the OH* chemiluminescence depends on the concentration of the electronically excited OH* radicals where the superscript * denotes the electronically excited state of the OH radical. These radicals are formed by chemical reactions predominantly via $H+O+M \rightarrow OH^*+M$ [25]. Due to the short lifetime of OH* at the investigated pressures, the chemiluminescence originates only from the reaction zone. Therefore, this technique yields information about the size and position of the flame zone [26]. The signal was detected through a side window with a charge coupled device (CCD) camera (gate time of 40 μs) and an achromatic 100 mm UV lens (f/2, Halle Nachf.) with an UV interference filter ($\lambda=312\pm20$ nm). Because the signal originated from the entire combustion chamber, the camera recorded an image that is spatially integrated in the depth (line of sight).

3.5 Particle Image Velocimetry (PIV). In this work, two-component velocity fields in two planar sections through the combustion chamber were measured using particle image velocimetry (PIV). The PIV method determines local fluid velocities by analyzing the displacement of small seeding particles during a short time span. For measuring this displacement, the particles in a planar section are illuminated twice by a double-pulsed laser with an appropriate separation time. A schematic of the PIV setup is shown in Fig. 6. A dual-cavity flashlamp-pumped neodymium doped yttrium aluminum garnet (Nd:YAG) laser emits two pulses ($\lambda=532$ nm, pulse energy of 120 mJ, and pulse duration of 5 ns) that are triggered synchronously with the exposure gates of the PIV camera. Two cylindrical lenses ($f_1=-12.7$ mm, $f_2=200$ mm) convert the laser beam into a 140 mm wide laser sheet. Behind the cylindrical lenses, a spherical lens ($f_3=1000$ mm) focuses the sheet and reduces its planar divergence. A large bending mirror guides the laser sheet vertically through the combustion chamber. The lens of the PIV camera (focal length

Downloaded 15 Jan 2011 to 129.137.89.240. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

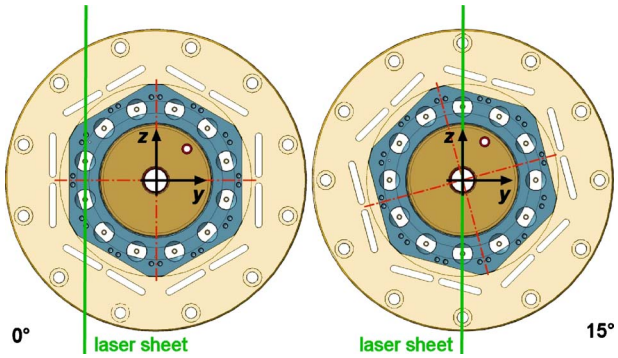

Fig. 7 Arrangement of burner and measurement plane for configurations of 0 deg and 15 deg

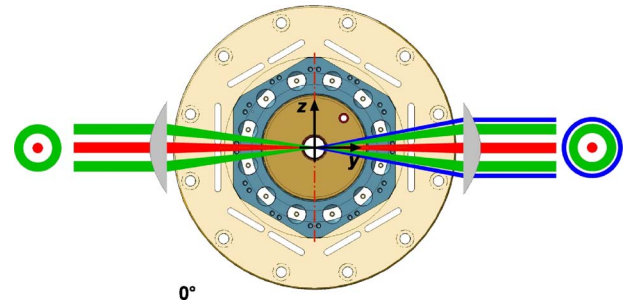

Fig. 8 Cross sectional view of the combustion chamber with the enhanced FLOX® burner. The two exciting CARS laser beams (pump laser; ring on the left side; Stokes laser: spot on the left side) were focused with a lens to the probe volume. The generated CARS signal (narrow outer ring on the right side) is collected with a second lens.

of 50 mm, F5.6) is covered with a narrow band interference filter (532 nm), reducing the influence of broadband light sources such as the flame. An air flow seeded with titanium dioxide particles (nominal diameter of 1 μm) is generated by a fluidized bed seeder and then fed into the main air flow. The seeded air flow is only applied during the PIV measurements because it reduces window transmittance (see below). The recording rate of the PIV system is 5 double frames/s. The separation time between the two laser pulses was chosen between 4 μs and 8 μs depending on the operating condition.

The PIV measurements are encumbered by the harsh conditions of the high pressure test rig. Temporal fluctuations of temperature and density in the combustion chamber degrade planeness and homogeneity of the laser sheet. As temperatures in the combustion chamber rise above 1600°C, the windows of the chamber have to sustain very high heat loads. The glass begins to soften, and its surface becomes rough and increasingly opaque. Furthermore, during the PIV measurements the window staining was significantly increased by seeding particles adhering to the quartz glass. This is a major problem for the PIV technique because the stray light formed at the window surface has the same wavelength as the light scattered by the particles. The increasing stray light due to window coating leads to a continuously decreasing quality of the particle images. After a total of 5 min of seeded air flow, the stray light was too strong to allow further imaging of seed particles. The test rig was then shut down, and the windows were cleaned or replaced.

As a first step, particle images were preprocessed in order to remove stationary patterns caused by light reflections or impurities on the windows. Velocity fields were determined from the preprocessed images using commercial software (DAVIS 7). For good quality images (low stray light), instantaneous velocity fields were calculated using an interrogation window size of 64 $\times$64 pixel (corresponding to a spatial resolution of 7×7 mm^2) and 50% overlap. Below a certain image quality, the cross-correlations calculated for a single image pair were too erroneous to determine the correct instantaneous velocity field. For the determination of average velocities, this problem was bypassed by applying a method called "ensemble correlation" or "sum of correlation" (SOC). This method first sums up the cross-correlations of all single shots and calculates the vector field afterward [27,28]. The so calculated velocity fields are practically equivalent to the temporal average and will be referred to as the average flow field in the following.

Velocity fields were measured in two planar sections through the combustion chamber with different azimuthal orientations of the burner (Fig. 7). The coordinates x, y, and z are defined in the streamwise direction (horizontal), in the horizontal direction perpendicular to x, and in the vertical direction, respectively (Figs. 7 and 9(a)). The vertical direction is also named as the height $h=z$. The position $x=y=z=0$ mm is located at the intersection point of the central axis of the burner and the plane with 12 mm distance to the burner front plate. For a configuration of 0 deg, the measurement plane is located downstream of two neighboring nozzles. Due to strong internal reflections of laser light in the upper part of the area ($z>0$), velocities were determined only in the lower part (-25 mm$<z<-5$ mm). The range in streamwise direction was 0 mm$<x<$140 mm. For a configuration of 15 deg, the burner was rotated by 15 deg around the central axis of the burner, and the measurement plane is located in the center of the combustion chamber. Here, velocities were determined for -5 mm$<z<$20 mm and -5 mm$<x<$140 mm.

3.6 Coherent Anti-Stokes Raman Scattering (CARS). CARS spectroscopy has been established in the past as a laser-based, nonintrusive temperature measuring method for combustion processes in a practical environment using mainly N_2 as the indicator molecule [29,30]. The CARS signal is coherently generated and emitted in a laserlike beam. Using a broadband CARS technique allows us to detect an entire N_2 Q-branch spectrum in a snapshot generated with a single laser pulse lasting about 10 ns. Because the spectral distribution of the Q-branch displays the population of the rotational and vibrational states, the temperature can be determined directly from the spectrum without calibration. Usually, the temperature is derived from the comparison of the shape of computed and experimental N_2 Q-branch spectra. Single shot measurements can be carried out with the repetition rate of the laser (10 Hz). Typically, 1200 single shot measurements were performed at each measuring location. The individual, instantaneous temperature values measured at the same location can be represented in the form of a histogram, from which temperature fluctuations can be read. In addition, statistical information as mean temperature T_{mean} and most probable temperature T_{mopro} were evaluated. With this pointwise measuring technique, different temperature profiles were recorded.

The mobile CARS apparatus of the DLR Institute of Combustion Technology has been designed for single shot N_2 thermometry in an industrial environment. It consists of a pulsed Nd:YAG laser operated at $\lambda=532$ nm (termed pump laser) with a repetition rate of 10 Hz and a broadband dye laser at $\lambda=607$ nm (termed Stokes laser). The USED-CARS arrangement was employed to overlap the two exciting laser beams at the measuring location (compare Fig. 8). In the overlapping volume, the CARS signal is generated, and it thus represents the size of the measuring volume and the spatial resolution. The spatial resolution is defined as the length of the measuring volume in beam direction in which 95% of the CARS signal is generated. From a measurement without flame, the spatial resolution was found to be $L_{95\%}=4.2$ mm. The diameter of the probe volume was approximately 300 μm.

The signal receiving optics at the opposite side of the combustion chamber separated the CARS signal from the laser beams and

Downloaded 15 Jan 2011 to 129.137.89.240. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

focused it into an optical fiber with a core diameter of 600 μm and a length of 20 m. At the end of the fiber, the signal was coupled into a spectrometer and was detected with an intensified gateable diode array camera.

After correcting each single shot spectrum, a contour fit of theoretical spectra to experimental data was performed to deduce the single shot temperatures. To accelerate the data analysis, a library of theoretical spectra is calculated for the expected temperature range with temperatures in steps of 50 K. The start temperature for the library-based least-squares fitting routine is given by a quick fit method (e.g., using the ratio of the integrated intensity of the fundamental and the first excited vibrational band). A more detailed description of the data evaluation procedure is found in Ref. [31]. The temperature accuracy of the mobile CARS system was tested in an isothermal furnace for single shot measurements and is in the range of $\pm 3\%$. Due to the high turbulence in the high pressure flames, investigated beam steering effects caused by fluctuating density gradients led to a strong degradation of the alignment of the overlapping laser beams and decreased the CARS signal intensities considerably. Therefore, the accuracy for the experiments presented in this paper was approximately $\pm 5\%$ for the mean temperatures. In some measuring positions, strong temperature fluctuations and especially temperature gradients occurred. The spatial resolution ($L_{95\%}=4.2$ mm without flame) is impaired by the fluctuation and is expected to be in the range of about 5–10 mm depending on the magnitude of the beam steering effects.

If cold and hot regions are simultaneously present in the measuring volume, the single shot CARS spectrum consists of superimposed spectral parts of different temperatures (mixed spectrum). This problem in flames with strong temperature gradients is discussed in the literature as mixed spectra [32] or as spatial averaging [33]. Due to the nonlinearity of the CARS process, the resulting evaluated temperature from the single shot CARS spectrum does not represent the mean mass- or volume-averaged temperature. The evaluated temperature is strongly biased to low temperatures; e.g., it is in the range of temperatures in the cold region even if the cold volume is only about 20% of the total probe volume [33,34].

The temperature distributions were measured in a vertical plane through the axis of the burner. The positions could be changed by moving the optics on both sides of the combustion chamber with computer controlled translation stages. The uncertainty in positioning the measuring volume was estimated to be less than ± 1 mm, which depends mainly on the thermal expansion of the test rig. The position $x=0$ mm was checked with the laser beams during operation.

4 Results

4.1 Flame Position. Images of the line of sight integrated OH^* chemiluminescence signals taken at different operating conditions are shown in Fig. 9. The pressure $p=7$ bars, the nozzle exit velocity $v_{\text{nozzle}}=90$ m/s, and the air preheat temperature $T=673$ K were kept constant. In Fig. 9(*c*), natural gas with $\lambda=1.6$ is used as fuel. Figure 9(*d*) displays two results for natural gas+50 vol % hydrogen, with $\lambda=1.6$ and $\lambda=2.0$, respectively.

As mentioned before, the chemiluminescence is spatially integrated in depth; i.e., all nozzles in the combustor contribute to the recorded intensity. Figures 9(*b*) and 9(*c*) illustrate the positions of the nozzles and their horizontal projections relating to the signal intensity. During the operation of the combustor, the quartz glasses are stained due to the high temperatures near the walls. The change in their transparency is inhomogeneous. As a consequence, this window clouding causes gaps in the recorded intensity of the chemiluminescence.

The flame position and size in the enhanced FLOX® combustor are different for natural gas and natural gas with 50 vol % hydrogen. The flames for hydrogen containing fuel mixtures are shorter, and their distance to the burner nozzles is smaller due to the

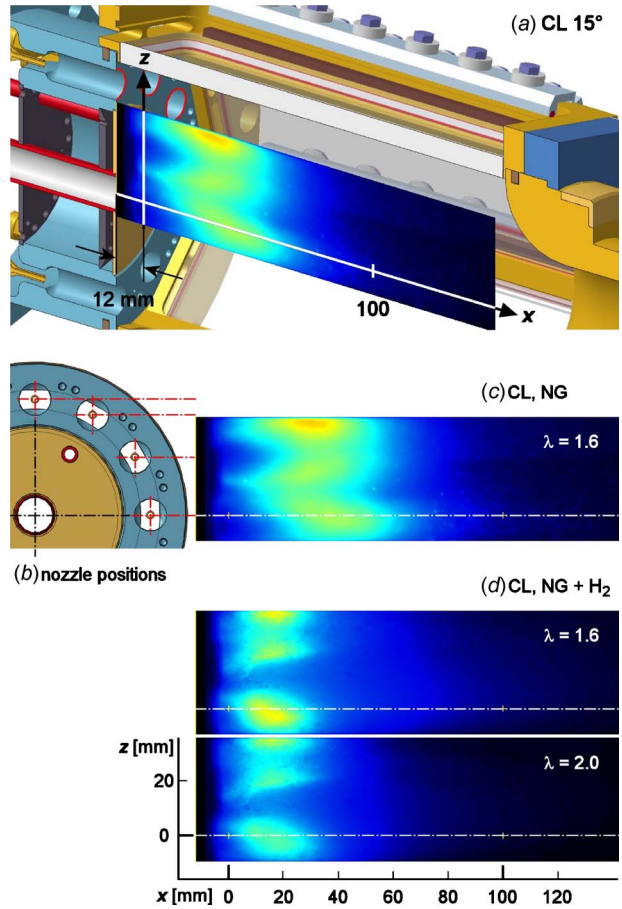

Fig. 9 Flame position and size for different fuels and air equivalence ratios for enhanced FLOX® burner, version 2A. (*a*) Configuration of the combustor: 15 deg, indication of the size and the position of the chemiluminescence image plane; (*b*) front view of the burner, projection of the positions of the nozzles. Measurements for (*c*) natural gas and (*d*) natural gas +50 vol % hydrogen.

shorter ignition delay time of hydrogen compared with natural gas. The higher reactivity of hydrogen leads to combustion under locally smaller air equivalence ratios λ_{local} than the air equivalence ratio of the global mixture λ_{global} (the chemical time scale decreases, relating to the time scale of the mixing in the combustion chamber). In all cases, the flames lift off the burner front plate, and flashback is not detected.

4.2 Pressure Dynamics. Two typical frequency spectra of the high pressure tests are given in Fig. 10. The burner was operated with a pressure of $p=7$ bars, an air preheat temperature of $T=673$ K, and natural gas as fuel. The curve in Fig. 10(*a*) corresponds to a nozzle exit velocity of $v_{\text{nozzle}}=120$ m/s; the line in the lower diagram corresponds to $v_{\text{nozzle}}=70$ m/s (Fig. 10(*b*)). The periodic peaks at 50 Hz, 150 Hz, etc., are caused by electronic interferences. For 120 m/s, the measured dynamic pressure amplitudes (rms value) stay below 10 mbars. For low nozzle exit velocities, a peak at 634 Hz can be observed. At $v_{\text{nozzle}}=70$ m/s, the amplitude rises to 20.7 mbars, which is equivalent to 2.96% of the pressure in the combustor. In general, the pulsation amplitudes are very low even if the nozzle exit velocity falls below the design point.

4.3 Flow Field. Computational results for the flow field are given in several drawings (Figs. 1, 11(*a*), 12(*b*), and 13) where streamlines represent the averaged global flow. A strong inner recirculation zone dominates the flow. The outer recirculation

Downloaded 15 Jan 2011 to 129.137.89.240. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

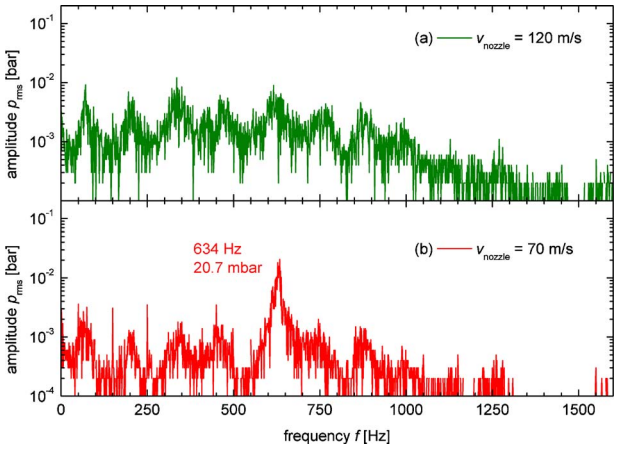

Fig. 10 Pressure dynamics for different nozzle exit velocities for enhanced FLOX® burner, version 2A, operated with natural gas

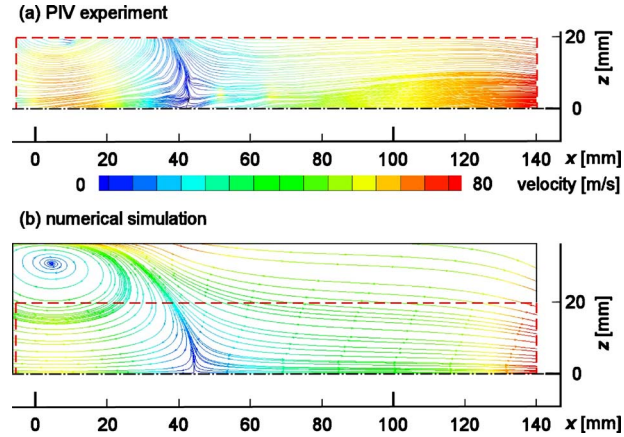

Fig. 12 Average flow field (streamline plot) for enhanced FLOX® burner, version 2A. Comparison between (*a*) PIV experimental data and (*b*) result from numerical simulation.

zone present in former versions of the FLOX® combustor (see Ref. [13]) is missing due to the increased pitch circle diameter of the jet nozzles.

Figure 11(*b*) shows four instantaneous velocity fields in the configuration of 15 deg of the enhanced FLOX® burner, version 2A, operated at a pressure $p=7$ bars, nozzle exit velocity $v_{\text{nozzle}}=90$ m/s, and air preheat temperature $T=673$ K with natural gas. The size and the position of the accessible area for the measurements are indicated by a dashed frame in Fig. 11(*a*). In the area $x<50$ mm, the velocity fields are characterized by an upstream recirculation that fluctuates strongly both in magnitude and in length. For $x>70$ mm, the flow is mostly stable and homogeneous in the positive x-direction, and the volumetric flow rate increases in the downstream direction. Stagnation points (axial velocity $u=0$) are most frequently found in the range 40 mm $<x<60$ mm. In the upper left corners, velocities are often flawed due to very low concentrations of seeding particles in this region.

Streamline plot images of the average flow field from the high pressure test and the numerical simulation for the same operating conditions are compared in Fig. 12. The experimental data shown are a result of the SOC method described in the previous section. The positions of the stagnation point and the slope of the tangent line match. The numerical simulation predicted slightly lower upstream velocities in the recirculation region.

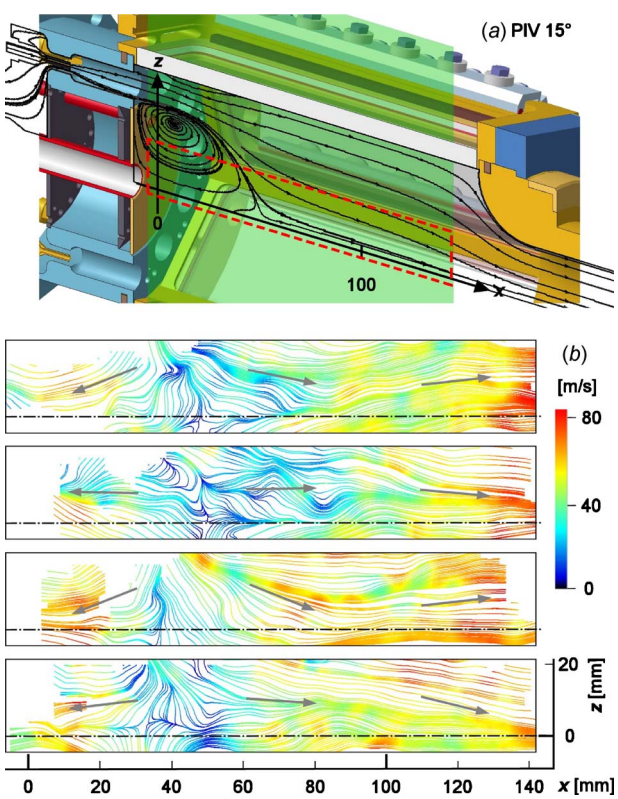

Fig. 11 Flow field measurements with PIV for enhanced FLOX® burner, version 2A. (*a*) Configuration of the combustor: 15 deg, indication of the size and the position of the PIV laser sheet and image plane; (*b*) instantaneous flow fields (streamline plots).

4.4 Temperatures. Figure 13 shows results from time-averaged VLES simulations of the HiPerMix® burner, version 2A, operated at a pressure $p=7$ bars, a jet nozzle exit velocity $v_{\text{nozzle}}=90$ m/s, and an air preheat temperature $T=673$ K with natural gas (Fig. 13(*a*)) and natural gas with 50 vol % hydrogen (Fig. 13(*b*)) as fuel. The calculated adiabatic flame temperature is $T_{\text{ad}}\approx 2000$ K for both cases.

The temperature fields for both cases are not completely homo-

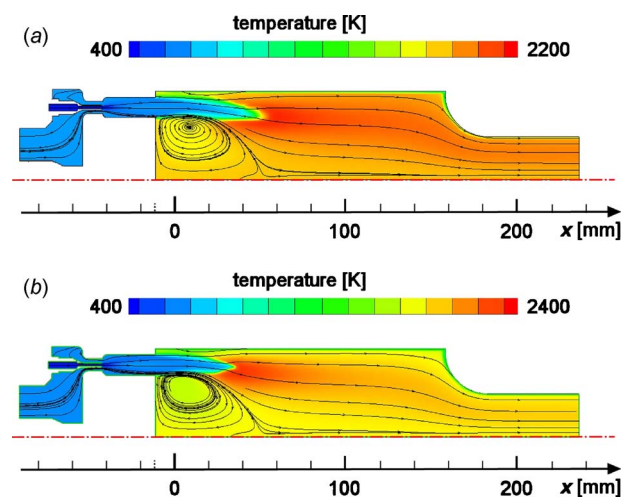

Fig. 13 Results from numerical simulation: temperature and flow fields for enhanced FLOX® burner, version 2A, operated (*a*) with natural gas and (*b*) with natural gas+50 vol % hydrogen, respectively

Downloaded 15 Jan 2011 to 129.137.89.240. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

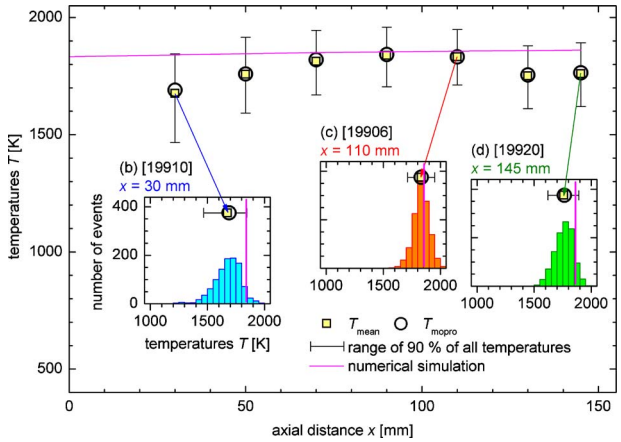

Fig. 14 Temperature measurements with CARS and comparison with computational values. Temperature profile on the combustor axis. Small inlaid frames: statistical temperature information for selected measurement positions (histograms with number of events versus temperatures, all inlaid frames scaled equally).

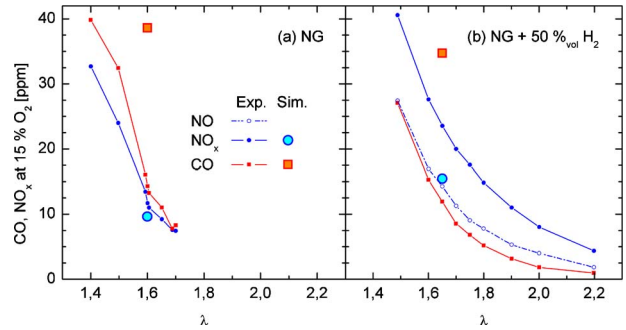

Fig. 15 Exhaust gas analysis data for enhanced FLOX® burner, version 2A: (*a*) natural gas and (*b*) natural gas +50 vol % hydrogen. Computational results indicated by big symbols, squared for CO, and circular for NO/NO_x, respectively.

geneous, and the maximum temperature exceeds the adiabatic flame temperature. The highest local temperatures are $T_{max}=2130$ K for natural gas and $T_{max}=2305$ K for hydrogen admixture, respectively.

During the high pressure tests, temperature measurements with CARS were carried out. The results for the operating conditions $p=7$ bars, $v_{nozzle}=90$ m/s, $T=673$ K, and $\lambda=1.6$ with natural gas as fuel are displayed in Fig. 14. For each measuring position, the mean temperature T_{mean} (squared symbol), the most probable temperature T_{mopro} (circular symbol), and the interval that comprises 90% of all measured temperatures at this position are given. For asymmetric temperature distributions, the most probable temperature T_{mopro} and the 90% interval of all temperatures are more suitable to represent the temperature fluctuations. Histograms were generated for each measuring position based on the single shot temperature values. Some temperature histograms are shown in Fig. 14 for selected CARS measuring positions.

The number N of evaluated spectra is smaller than the number of recorded spectra (1200) because some spectra were rejected due to a too low intensity (bad signal-to-noise ratio), a too high intensity (saturation of the analog to digital (AD) converter), or a too large sum of residuals (bad quality of the spectrum). Spectra containing a high background level (nonresonant CARS signal or optical breakdown producing radiating plasma initiated by intense laser light in the focal region) were also rejected due to the undefined influence on the accuracy of the resulting temperature value. In most situations, the number of rejected spectra is small with respect to the 1200 recorded ones.

The instantaneous temperature data indicate different degrees of fluctuations in the flame. At $x=30$ mm ($z=y=0$ mm), a position in the inner recirculation zone, the temperature distribution is expanded compared with positions downstream the stagnation point. The location $x=110$ mm is characterized by the smallest interval of 90% of all temperatures. Beyond this point, the temperature unsteadiness increases again.

The solid line given in Fig. 14 is a result of the numerical simulation, as shown in the upper half of Fig. 13. In the central region of the combustor, the conformance of experimental and computational data values is very good. At locations ahead the stagnation point and toward the exhaust gas nozzle, the predicted temperature from numerical simulation is too high. While the deviation for $x=30$ mm is significant, the discrepancy for all other measurement positions is smaller than the measuring inaccuracy of the CARS method in a turbulent combustor (±5%, i.e., ±93 K at this temperature level). As stated in a previous section, CARS tends to overestimate cold temperatures due to the nonlinearity of the CARS process. However, all temperature values match within the fluctuation ranges.

4.5 Exhaust Gas Analysis. The conventional exhaust gas analysis system of the high pressure test rig was connected to a suction probe behind the combustion chamber. Its inlet is located near the center axis of the combustor (because of the tight narrowing of the exhaust gas nozzle the radial position had no influence on the measured values). The concentrations of the pollutants CO and NO/NO_x in this work are converted to dry exhaust gas related to 15% O_2 concentration.

Experimental and computational results for the version 2A of the HiPerMix® burner are shown in Fig. 15. The burner was operated at $p=7$ bars, $v_{nozzle}=90$ m/s, and $T=673$ K. The fuel mass flow of natural gas (Fig. 15(*a*)) and of natural gas +50 vol % hydrogen (Fig. 15(*b*)) was varied.

For natural gas (Fig. 15(*a*)), the NO_x curve goes below the limit of 10 ppm at $\lambda=1.6$ (adiabatic flame temperature $T_{ad}=1996$ K) as predicted from numerical calculations. In the case of the natural gas+hydrogen mixture, the computational values at $\lambda=1.65$ for the concentration of NO at the combustor exit is 15.4 ppm, and that of NO_2 is 0.05 ppm. Here, the simulated NO values are consistent with the measured NO concentrations, while the concentration of NO_2 is higher as forecasted.

For an almost perfectly premixed combustion, Leonard and Stegmaier [35] experimentally investigated the NO_x formation subject to the adiabatic flame temperature in the range of 1700–1900 K. From this work, we find by extrapolation concentrations of NO_x between 12 ppm and 19 ppm depending on the residence time between 2 ms and 100 ms. In fact, if the fuel/air mixture is assumed to be perfectly premixed, the simulation predicts about 13 ppm NO_x emission. However, in the present case, the emission value of 10 ppm is lower despite the fact that the mixture is obviously not perfect (see Fig. 13). The reason for this is seen in Fig. 1. We observe that the flame is largely detached from the burner. NO_x is predominantly generated at the tip of the flame and directly transported to the combustor exit. The residence time between these two points is about 2 ms. In the case of a perfectly premixed combustion, the simulation reveals that the flame extends more upstream between jet and recirculation, and consequently more NO_x is generated in the recirculation zone due to the essentially larger residence times.

Figure 16 shows nitrogen oxide (NO_x) values measured at the combustor exhaust gas nozzle for different fuel mixtures. The air equivalence ratio was varied; the curves are printed for the corresponding adiabatic flame temperature values, which are deduced from equilibrium calculations and do not consider possible heat

Downloaded 15 Jan 2011 to 129.137.89.240. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

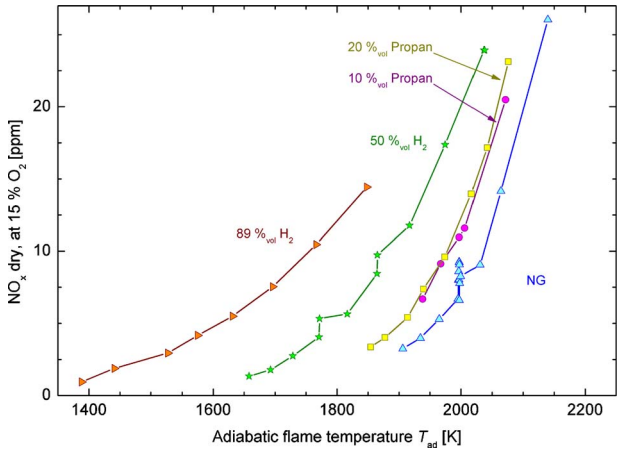

Fig. 16 Measured NO_x values for enhanced FLOX® burner, version 2D, for different fuel mixtures (natural gas, natural gas + propane, natural gas + hydrogen)

loss in the combustion chamber. The constant operating conditions were pressure $p=7$ bars, $v_{\text{nozzle}}=133$ m/s, and air preheat temperature $T=673$ K.

The enhanced FLOX® combustor, version 2D, could be operated with natural gas (methane content 95–98 vol %), natural gas + propane up to 20 vol % C_3H_8, and natural gas + hydrogen up to 100 vol % H_2. NO_x values below 10 ppm could be achieved in all cases. The combustor was optimized for natural gas and combined <10 ppm NO_x performance with high adiabatic flame temperature operation (2000 K).

5 Conclusion

Modern gas turbine combustors operate at high specific power densities. Next generation gas turbines will again raise the turbine inlet temperature. In previous work, it was demonstrated that the FLOX® burner can be operated with different fuel compositions, especially with hydrogen content [13]. The challenge in the development process of an enhanced FLOX® burner is the combination of these three attributes. The novel burner concept based on the FLOX® principle should merge specific power densities of $P_A > 10$ MW/m^2 bar, adiabatic flame temperatures of $T_{ad}=2000$ K, and low NO_x emission values.

The design process is supported by two essential parts, numerical simulations that can predict emission values by the usage of detailed chemistry models and high pressure tests combined with revealing combustion diagnostic techniques.

Identifying the influence of the fuel distribution in the combustion air prior to the combustion chamber is a key issue. The HiPerMix® principle was applied to the FLOX® nozzles to get a reliable control mechanism for the mixing behavior.

The enhanced FLOX® burner, also called HiPerMix® burner, could be operated at the high pressure test rig HBK-S with pressures up to 7 bars and nozzle exit velocities up to 150 m/s. Experiments with natural gas, natural gas + hydrogen, and natural gas + propane were carried out. Stable operation is achieved for fuel blends up to 100 vol % hydrogen and 20 vol % propane. This demonstrates the high fuel flexibility of the burner concept, which makes it attractive to meet requests for next generation gas turbines.

Flame positions, pressure dynamics, flow fields, temperatures, and exhaust gas compositions were measured. The experimental values are used as code and model validation data besides their usage for a deeper understanding of the combustion behavior. Predicted quantities are in close match with experimental results.

Low NO emissions are attained if the local temperature values do not exceed the adiabatic flame temperature of the global mixture. Therefore, the enhanced FLOX® burner development aims at an almost homogeneous temperature field. Recent numerical simulations and CARS temperature measurements show a remaining inhomogeneity in the temperature field of the HiPerMix® burner. This is due to a compromise between mixing performance and burner pressure loss. However, combustor operation with NO_x concentrations below 10 ppm could already be achieved in combination with adiabatic flame temperatures of about 2000 K and specific power densities of more than 10 MW/m^2 bar.

Acknowledgment

The authors would like to thank Uwe Prestel and Karl-Heinz Ferst for the installation of the combustor and the operation of the test rig and Manfred Kapernaum, Oliver Kislat, Sami Tanios, and Christian Eberle for the assistance during the experiments and the data evaluation.

The investigations were conducted as part of the joint research program COORETEC-turbo in the frame of AG Turbo. The work was supported by the Bundesministerium für Wirtschaft und Technologie (BMWi) as per resolution of the German Federal Parliament under Grant No. 0327715D. The authors gratefully acknowledge AG Turbo and Siemens AG Energy Sector for their support and permission to publish this paper.

Nomenclature

$\varnothing_{\text{nozzle}}$ = nozzle diameter (mm)
v_{nozzle} = nozzle exit velocity (m/s)
p = pressure (bar)
λ = air equivalence ratio
λ_{local} = local air equivalence ratio
λ_{global} = global air equivalence ratio
T = air preheat temperature (K)
P = thermal power (MW)
P_A = thermal power density per area (MW/m^2 bar)
P_V = thermal power density per volume (MW/m^3 bar)
T_{ad} = adiabatic flame temperature (K)
T_{max} = maximum local temperature (K)
Q = mass flow (g/s)
Q_{air} = air mass flow (g/s)
Q_{NG} = natural gas mass flow (g/s)
$f(H_2)$ = fuel mixture content, hydrogen H_2+NG (vol %)
$f(C_3H_8)$ = fuel mixture content, propane C_3H_8+NG (vol %)
λ = wavelength (nm)
f = focal length (mm)
x,y,z,h = coordinates (mm)
T_{mean} = mean temperature (K)
T_{mopro} = most probable temperature (K)
$L_{95\%}$ = spatial resolution (mm)
N = number of evaluated spectra

References

[1] Krebs, W., Hellat, J., and Eroglu, A., 2003, "Technische Verbrennungssysteme," *Stationäre Gasturbinen*, C. Lechner and J. Seume, eds., Springer–Verlag, New York.
[2] Correa, S. M., 1993, "A Review of NO_x Formation Under Gas-Turbine Combustion Conditions," Combust. Sci. Technol., **87**, pp. 329–362.
[3] Lefebvre, A. H., 1999, *Gas Turbine Combustion*, Taylor & Francis, Philadelphia.
[4] Dowling, A. P., and Hubbard, S., 2000, "Instability in Lean Premixed Combustors," Proc. Inst. Mech. Eng., Part A, **214**, pp. 317–332.
[5] Candel, S., 2002, "Combustion Dynamics and Control: Progress and Challenges," Proc. Combust. Inst., **29**, pp. 1–28.
[6] Kröner, M., Fritz, J., and Sattelmayer, T., 2002, "Flashback Limits for Combustion Induced Vortex Breakdown in a Swirl Burner," ASME Paper No. GT2002-30075.
[7] Lieuwen, T., 2003, "Combustion Driven Oscillations in Gas Turbines," Turbomachinery International, **44**, pp. 16–19.

Downloaded 15 Jan 2011 to 129.137.89.240. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm

[8] FLOX® is a registered trademark of WS Wärmeprozesstechnik GmbH, Renningen, Germany.
[9] Wünning, J. A., and Wünning, J. G., 1992, "Burners for Flameless Oxidation With Low NO_x Formation Even at Maximum Air Preheat," Gaswärme International, **41**(10), pp. 438–444.
[10] Wünning, J. A., and Wünning, J. G., 1997, "Flameless Oxidation to Reduce Thermal NO-Formation," Prog. Energy Combust. Sci., **23**, pp. 81–94.
[11] Cavaliere, A., and Joannon, M., 2004, "Mild Combustion," Prog. Energy Combust. Sci., **30**, pp. 329–366.
[12] Bobba, M. K., Gopalakrishnan, P., Seitzman, J. M., and Zinn, B. T., 2006, "Characteristics of Combustion Processes in a Stagnation Point Reverse Flow Combustor," ASME Paper No. GT2006-91217.
[13] Lückerath, R., Meier, W., and Aigner, M., 2008, "FLOX® Combustion at High Pressure With Different Fuel Compositions," ASME J. Eng. Gas Turbines Power, **130**(1), p. 011505.
[14] Schütz, H., Lückerath, R., Noll, B., and Aigner, M., 2007, "Complex Chemistry Simulation of FLOX®: Flameless Oxidation Combustion," Clean Air, International Journal of Energy for a Clean Environment, **8**(3), pp. 239–257.
[15] Schütz, H., Lückerath, R., Kretschmer, T., Noll, B., and Aigner, M., 2008, "Analysis of the Pollutant Formation in the FLOX® Combustion," ASME J. Eng. Gas Turbines Power, **130**(1), p. 011503.
[16] HiPerMix® is a registered trademark of the DLR (Deutsches Zentrum für Luft-und Raumfahrt e.V.), Köln.
[17] Hirt, C. W., Amsden, A. A., and Cook, J. L., 1974, "An Arbitrary Lagrangian-Eulerian Computing Method for All Flow Speeds," J. Comput. Phys., **14**, pp. 227–253.
[18] Pracht, W. E., 1975, "Calculating Three-Dimensional Fluid Flows at All Speeds With an Eulerian-Lagrangian Computing Mesh," J. Comput. Phys., **17**, pp. 132–159.
[19] Amsden, A. A., O'Rourke, P. J., and Butler, T. D., 1989, "KIVA-II: A Computer Program for Chemically Reactive Flows With Sprays," Report No. LA-11560-MS.
[20] Speziale, C. G., 1998, "Turbulence Modeling for Time-Dependent RANS and VLES: A Review," AIAA J., **36**, pp. 173–184.
[21] Nordin, P. A. N., 1995, "Complex Chemistry Modelling of Diesel Sprays," Ph.D. thesis, Chalmers University of Technology, Göteborg.
[22] Byggstøyl, S., and Magnussen, B. F., 1985, "A Model for Flame Extinction in Turbulent Flow," Fourth Symposium on Turbulent Shear Flows, pp. 381–395.
[23] Schütz, H., and Schmitz, G., 2009, European Patent No. PCT/EP08/59744 and German Patent No. DE 10 2007 036 953 B3 2009.04.02.
[24] Fleck, J., Griebel, P., Steinberg, A., Stöhr, M., and Aigner, M., 2010, "Experimental Investigation of a Generic, Fuel Flexible Reheat Combustor at Gas Turbine Relevant Operating Conditions," ASME Paper No. GT2010-22722.
[25] Hall, J. M., and Petersen, E. L., 2006, "An Optimized Kinetics Model for OH Chemiluminescence at High Temperatures and Atmospheric Pressures," Int. J. Chem. Kinet., **38**, pp. 714–724.
[26] Lee, J. G., and Santavicca, D. A., 2003, "Experimental Diagnostics for the Study of Combustion Instabilities in Lean Premixed Combustors," J. Propul. Power, **19**, pp. 735–750.
[27] LaVision GmbH, 2005, DaVis Flowmaster Manual.
[28] Meinhart, C., Wereley, S., and Santiago, J., 2000, "A PIV Algorithm for Estimating Time-Averaged Velocity Fields," ASME J. Fluids Eng., **122**, pp. 285–289.
[29] Eckbreth, A. C., 1995, *Laser Diagnostics for Combustion Temperature and Species*, 2nd ed., Gordon and Breach, Amsterdam.
[30] Stricker, W., 2002, *Measurement of Temperature in Laboratory Flames and Practical Devices*, 2nd ed., Taylor & Francis, New York.
[31] Lückerath, R., Woyde, M., Meier, W., Stricker, W., Schnell, U., Magel, H. C., Görres, J., Spliethoff, H., and Maier, H., 1995, "Comparison of Coherent Anti-Stokes Raman-Scattering Thermometry With Thermocouple Measurements and Model Predictions in Both Natural-Gas and Coal-Dust Flames," Appl. Opt., **34**, pp. 3303–3312.
[32] Meier, W., Plath, I., and Stricker, W., 1991, "The Application of Single-Pulse CARS for Temperature Measurements in a Turbulent Stagnation Flame," Appl. Phys. B: Photophys. Laser Chem., **53**, pp. 339–346.
[33] Boquillon, J. P., Péalat, M., Bouchardy, P., Gollin, G., Magre, P., and Taran, J. P., 1988, "Spatial Averaging and Multiplex Coherent Anti-Stokes Raman Scattering Temperature-Measurement Error," Opt. Lett., **13**, pp. 722–724.
[34] Thumann, A., Seeger, T., and Leipertz, A., 1995, "Evaluation of Two Different Gas Temperatures and Their Volumetric Fraction From Broadband N_2 Coherent Anti-Stokes Raman Spectroscopy Spectra," Appl. Opt., **34**, pp. 3313–3317.
[35] Leonard, G., and Stegmaier, J., 1993, "Development of an Aeroderivative Gas Turbine Dry Low Emission Combustion System," ASME Paper No. 93-GT-288.

Downloaded 15 Jan 2011 to 129.137.89.240. Redistribution subject to ASME license or copyright; see http://www.asme.org/terms/Terms_Use.cfm